

## **README - Dataset Usage, Licensing & Privacy Notice**

### **Project Overview**

This project uses multiple publicly available musculoskeletal and bone fracture imaging datasets for machine learning research, fracture detection, localization, segmentation, and classification tasks.

The datasets are used solely for research, development, training, validation, and evaluation of AI/ML models related to fracture analysis from radiographic imagery.

### **Datasets Used**

#### **1. FracAtlas Dataset**

##### **Citation**

Abedeenn, Iftekharul; Rahman, Md. Ashiqur; Zohra Prottyasha, Fatema; Ahmed, Tasnim; Mohmud Chowdhury, Tareque; Shatabda, Swakkhar (2023).

*FracAtlas: A Dataset for Fracture Classification, Localization and Segmentation of Musculoskeletal Radiographs.*

figshare. Dataset.

DOI: 10.6084/m9.figshare.22363012.v6

##### **Sources**

- FracAtlas Paper (Nature Scientific Data); <https://www.nature.com/articles/s41597-023-02432-4>
- FracAtlas Dataset (figshare); [https://figshare.com/articles/dataset/The\\_dataset/22363012](https://figshare.com/articles/dataset/The_dataset/22363012)
- Direct Dataset Download; [https://figshare.com/articles/dataset/The\\_dataset/22363012](https://figshare.com/articles/dataset/The_dataset/22363012)

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- SaaS platforms
- AI/ML model training
- Redistribution
- Model fine-tuning
- Derivative works

### **Attribution Requirement**

Any use of this dataset must provide proper attribution to the original authors.

#### **Attribution:**

**This project uses the FracAtlas dataset licensed under CC BY 4.0.**

## **2. Human Bone Fractures Multi-modal Image Dataset (HBFMID)**

### **Citation**

Parvin, Shahnaj (2024).

*Human Bone Fractures Multi-modal Image Dataset (HBFMID)*.

Mendeley Data, V1. DOI: 10.17632/xwfs6xbk47.1

### **Sources**

- HBFMID Dataset (Mendeley Data) ; <https://data.mendeley.com/datasets/xwfs6xbk47/1>
- Associated Research Paper; <https://link.springer.com/article/10.1007/s10489-024-05588-7>

### **Authors**

- Shahnaj Parvin
- Abdur Rahman

### **Intended Usage**

The dataset is used for:

- Bone fracture classification
- Real-time fracture detection research
- Multi-modal medical image analysis
- Deep learning experimentation

### **License Notice**

Users must review the dataset's official terms and licensing conditions directly from the dataset provider before redistribution or commercial deployment.

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## **Attribution**

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## **3. Bone Fracture Detection Using X-rays (Kaggle)**

### **Source**

- Kaggle Dataset

### **Citation**

Vuppala, A. S.

*Bone fracture detection using X-rays* [Dataset]. Kaggle.

### **Notice**

Usage rights, redistribution permissions, and commercial usage depend on Kaggle dataset terms and any additional restrictions imposed by the dataset author.

Users are responsible for independently verifying compliance before commercial deployment.

## **4. Bone Fracture Detection Dataset (Roboflow)**

### **Source**

- Roboflow Universe Dataset

### **Citation**

fyp. (2023).

*Bone fracture detection dataset.*

Roboflow Universe.

## **Notice**

This dataset remains subject to Roboflow Universe licensing terms and any restrictions imposed by the dataset uploader.

Users must independently verify permissions for commercial or production use.

## **Privacy Disclaimer**

### **Medical & Sensitive Data Disclaimer**

The datasets used in this project may contain medical imaging data such as radiographs/X-rays and MRIs. While these datasets are publicly distributed for research purposes, this project does not claim ownership of the original medical data.

The maintainers of this project:

- Do not collect personal patient data directly
- Do not intentionally store personally identifiable information (PII)
- Use datasets solely for machine learning research and software development purposes

Users and contributors are responsible for ensuring compliance with:

- Applicable privacy laws
- Medical data regulations
- Institutional review requirements
- Local jurisdictional restrictions

including but not limited to:

- GDPR
- HIPAA
- Regional medical data protection laws

## **Medical Disclaimer**

This software and associated machine learning models are intended for:

- Research
- Educational purposes
- Experimental AI development

They are **NOT** certified medical devices and must not be used as a substitute for professional medical diagnosis, treatment, or clinical decision-making.

No guarantees are made regarding:

- Diagnostic accuracy
- Clinical safety
- Medical reliability
- Regulatory compliance

Always consult qualified healthcare professionals for medical decisions.

### **Attribution Notice**

**This project includes datasets and derivative machine learning artifacts based on publicly available research datasets. All dataset copyrights remain with their respective authors and publishers.**

**Proper attribution is maintained in accordance with the original dataset licenses and publication requirements.**

#### **FracAtlas Attribution**

This project uses the FracAtlas dataset:

Abedeen et al. (2023), licensed under CC BY 4.0.

<https://doi.org/10.6084/m9.figshare.22363012.v6>

#### **HBFMID Attribution**

This project uses the Human Bone Fractures Multi-modal Image Dataset (HBFMID):

Parvin, S. (2024), licensed under CC BY 4.0.

<https://doi.org/10.17632/xwfs6xbk47.1>

### **Compliance Responsibility**

Anyone redistributing, deploying, commercializing, or modifying this project is responsible for independently verifying:

- Dataset licensing compatibility
- Medical AI regulatory requirements
- Privacy compliance obligations
- Commercial deployment legality in their jurisdiction

### **Third-Party Dataset Notice**

This project incorporates publicly available medical imaging datasets licensed by their respective authors.

Datasets used include:

1. FracAtlas

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2. Human Bone Fractures Multi-modal Image Dataset (HBFMID)

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